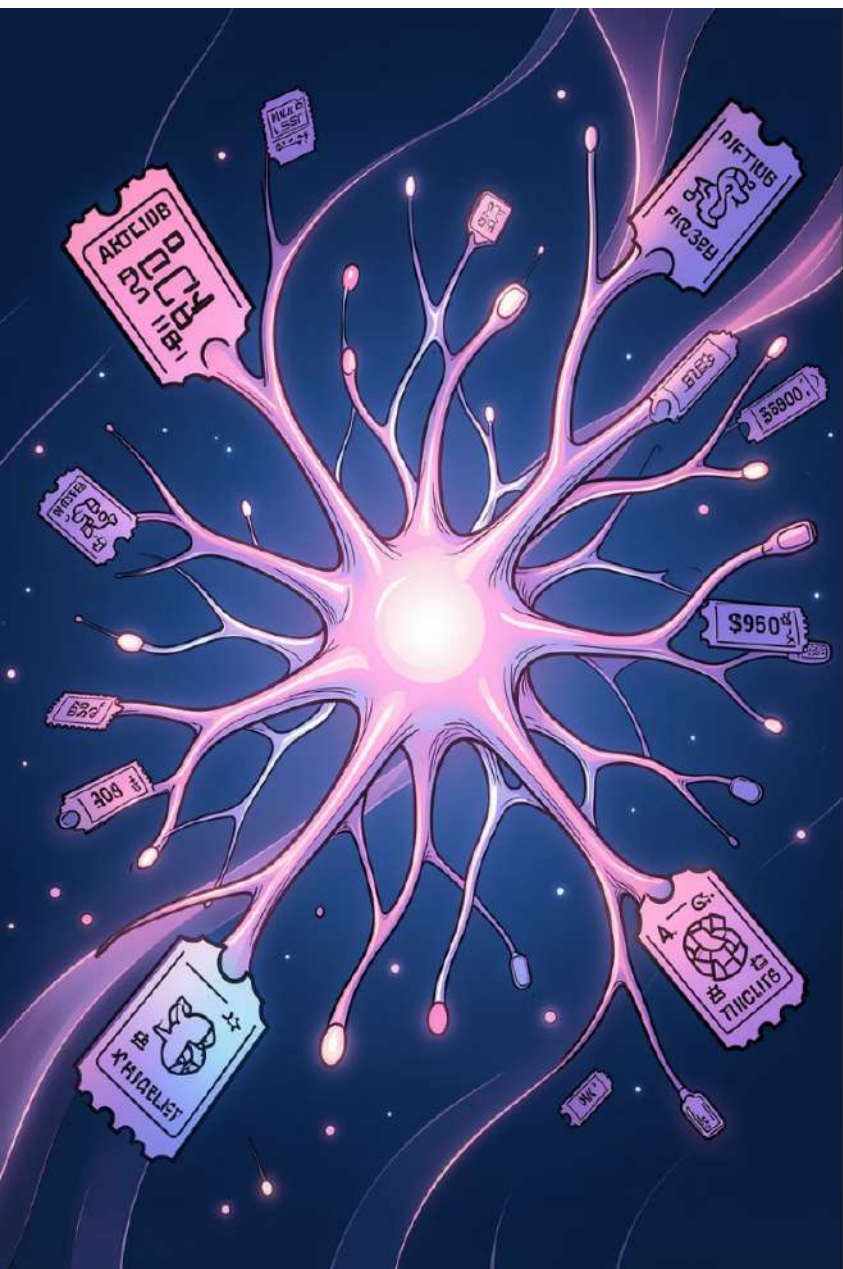




AI Duplicate Ticket Detector

A Django + React support desk that detects semantically similar tickets using AI embeddings and cosine similarity — showing users **"this might already be answered"** before they submit a duplicate.

DJANGO

REACT + VITE

AI EMBEDDINGS

COSINE SIMILARITY

The Problem It Solves



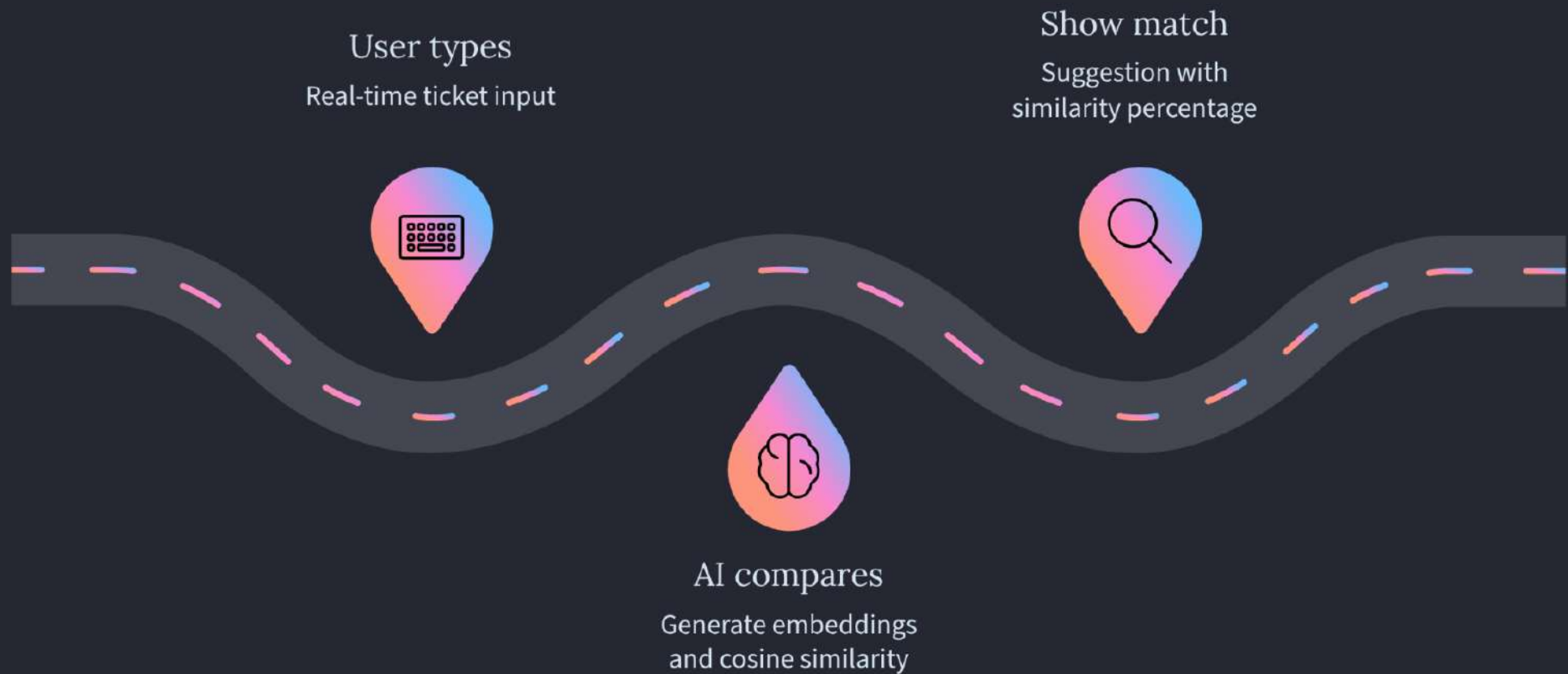
Why Duplicates Are Costly

Support teams waste time answering the same issue multiple times. Users get delayed responses, and ticket volume inflates artificially.

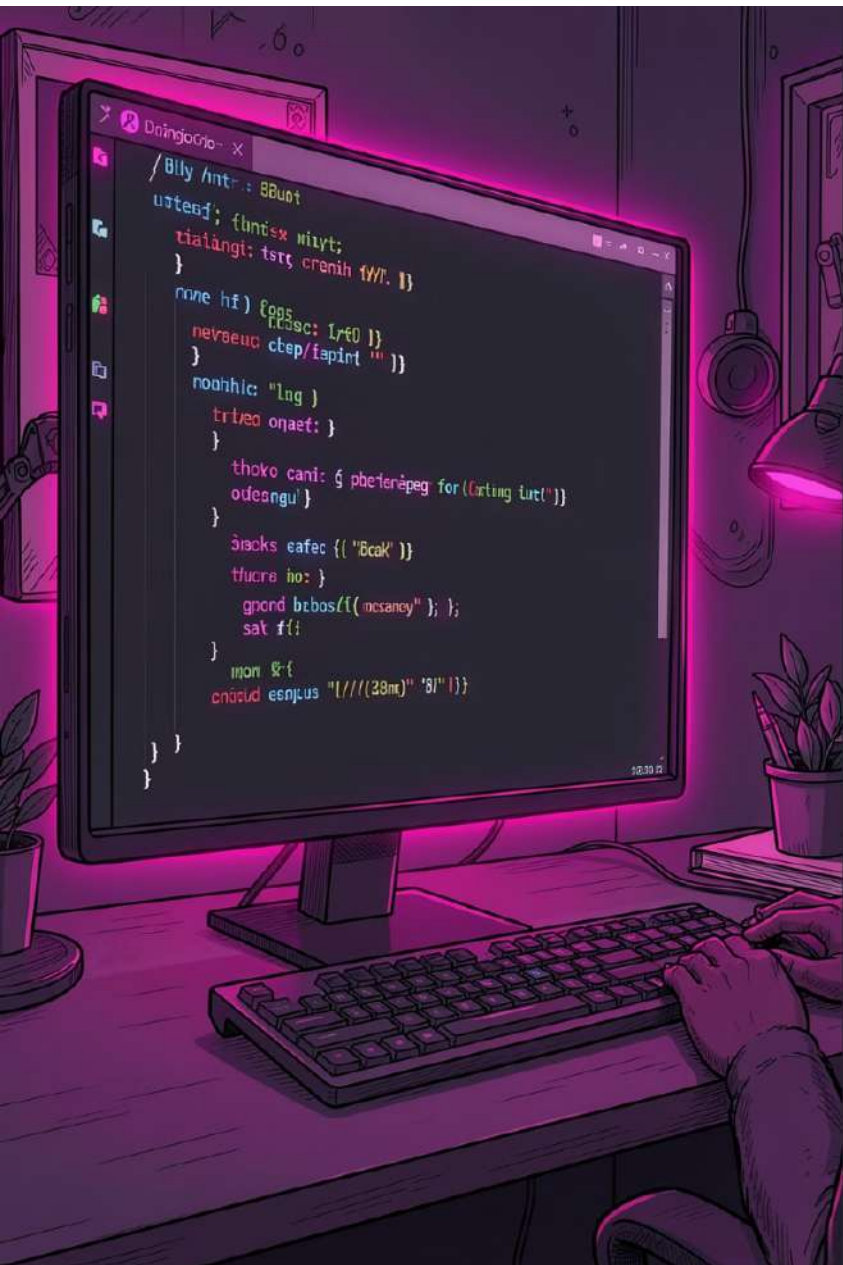
The AI Approach

By comparing the **semantic meaning** of ticket text — not just keywords — the system catches duplicates even when wording differs, surfacing existing answers before submission.

How It Works



The detection runs in real time as the user types. Once text input stops, embeddings are compared against existing tickets and a suggestion panel appears within about a second if a match is found above the similarity threshold.



Backend Architecture

Django Project

AI_support_ticket project with accounts app for auth and ticket app for ticket management and similarity logic.

Similarity Engine

similarity.py handles AI embeddings and cosine similarity scoring between ticket descriptions.

REST API

DRF serializers, JWT token auth via simplejwt, and structured API endpoints for tickets and authentication.

Frontend Architecture



React + Vite

Fast, modern SPA scaffolded with Vite. Components include TicketForm.jsx, AdminClusterDashboard.jsx, Login.jsx, and Register.jsx.



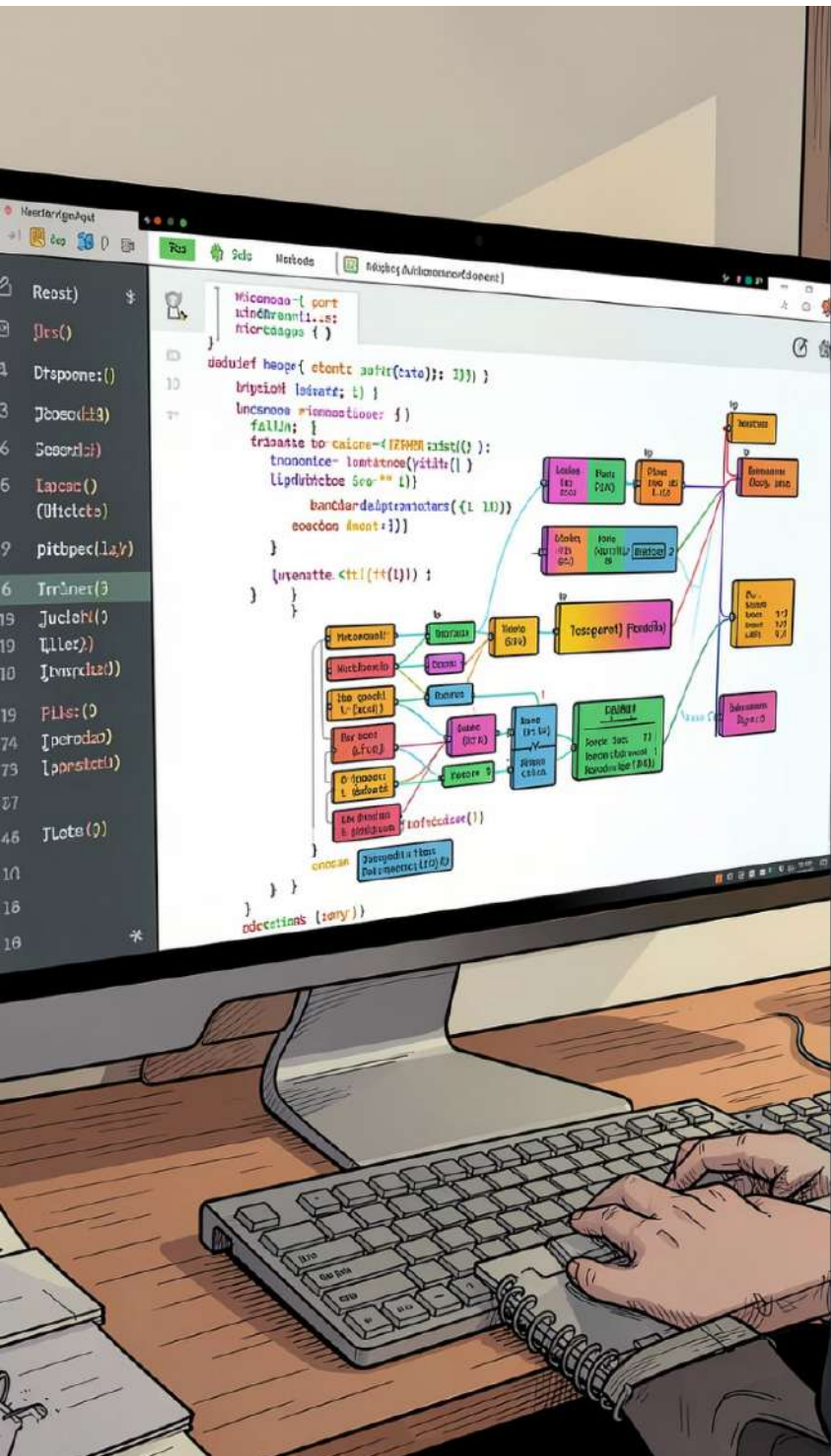
API Layer

Dedicated api/ folder with auth.js and tickets.js modules. JWT tokens are saved automatically after login.



Admin Dashboard

AdminClusterDashboard.jsx provides a visual overview of ticket clusters grouped by similarity.



Frontend Setup & Demo Walkthrough

Spin Up the Frontend

```
npm create vite@latest  
Support_desk  
cd Support_desk  
npm install  
npm run dev
```

Try the Demo

1. Register/login — tokens saved automatically
2. Create ticket #1: *"Payment failed after checkout... charged twice"*
3. Create similar ticket #2 — the yellow suggestion panel appears in ~1 second showing ticket #1 with a similarity percentage

